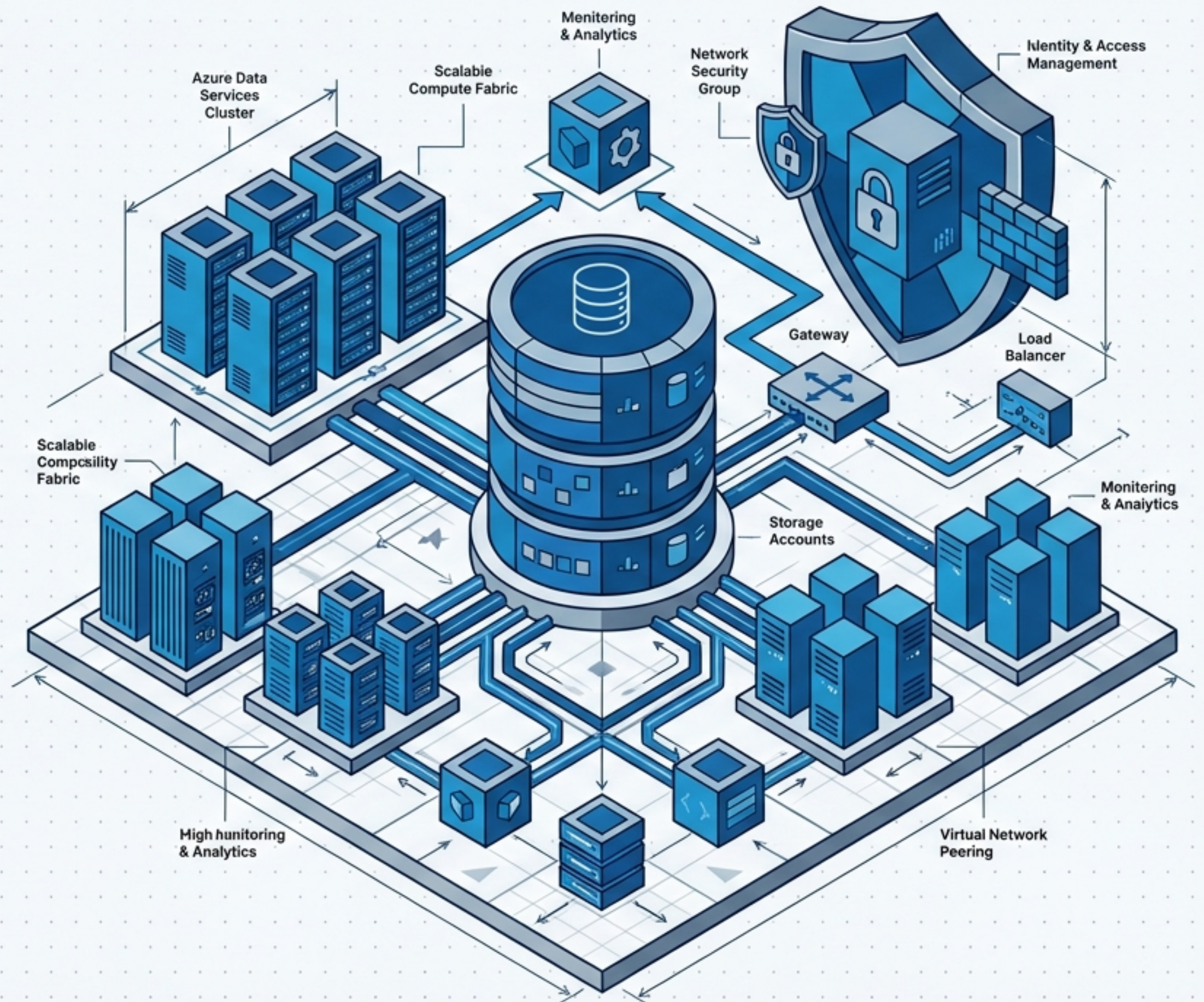


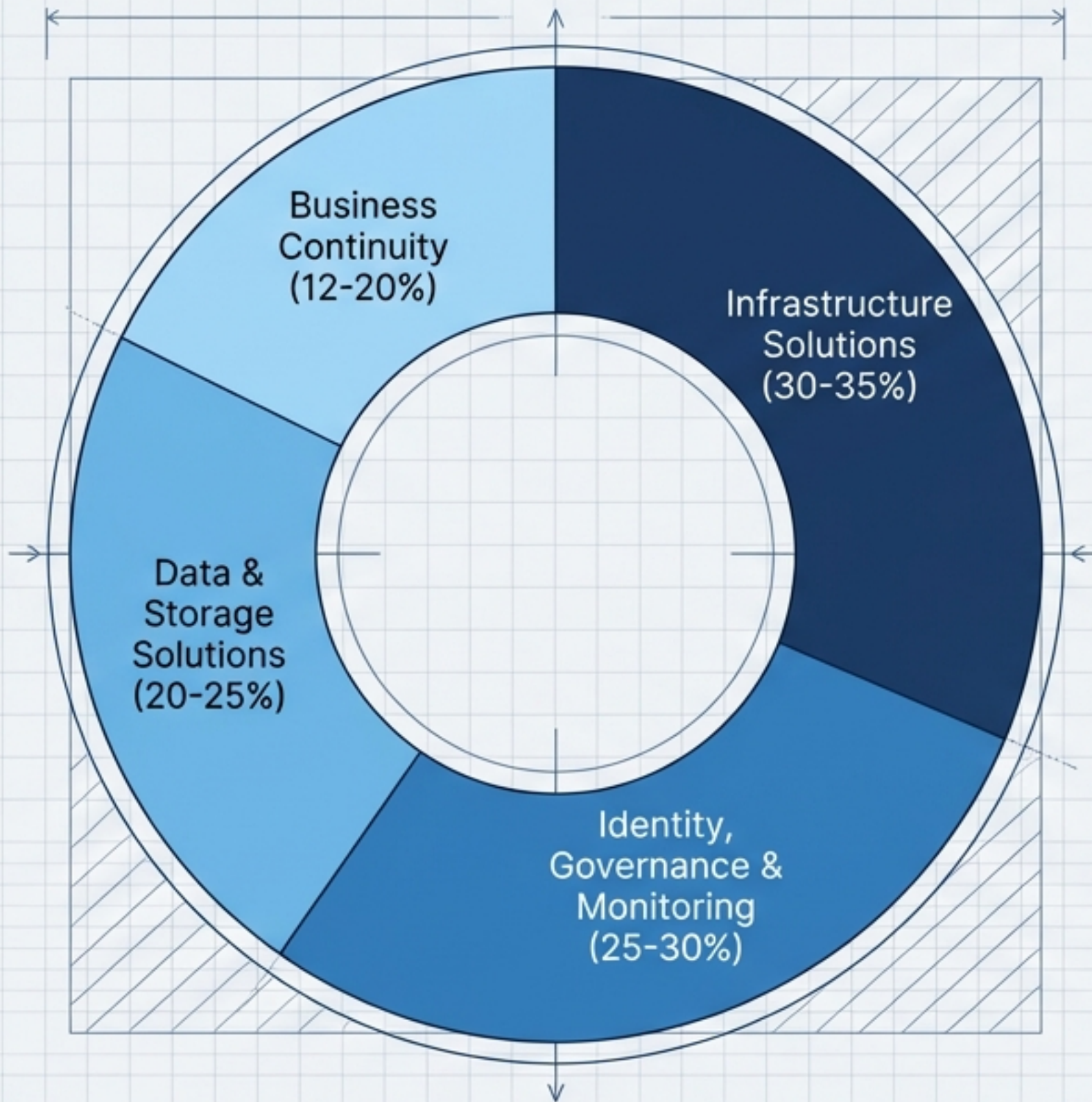
The AZ-305 Architect's Blueprint

A Visual Study Guide
for the Microsoft Azure
Solutions Architect
Expert Certification

Engineered for advanced cloud professionals



The 4 Exam Domains



Rules of Engagement

Time



120 Minutes

Load



40-60 Questions

Target



700 / 1000
to pass

Intel



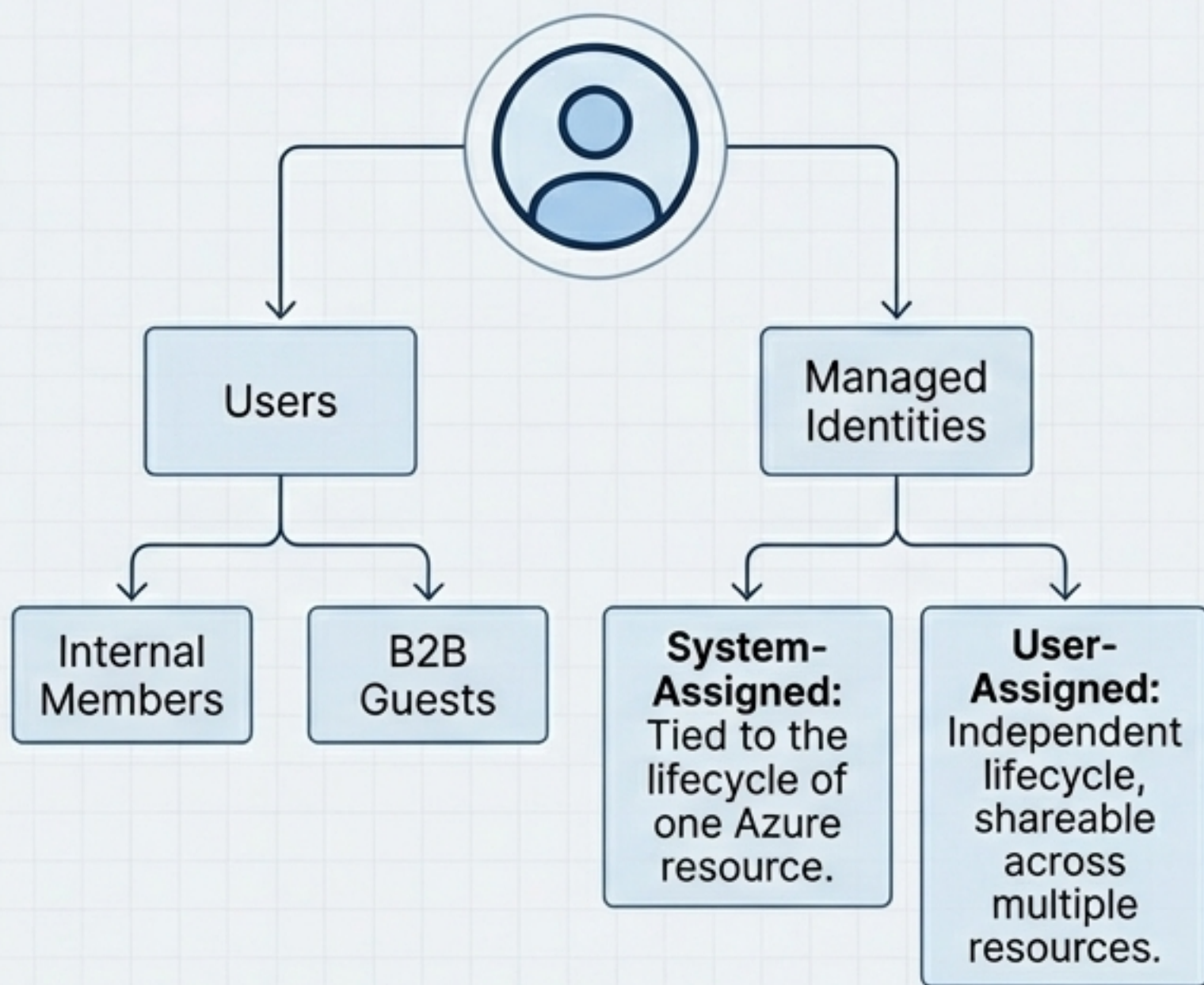
No penalty
for guessing



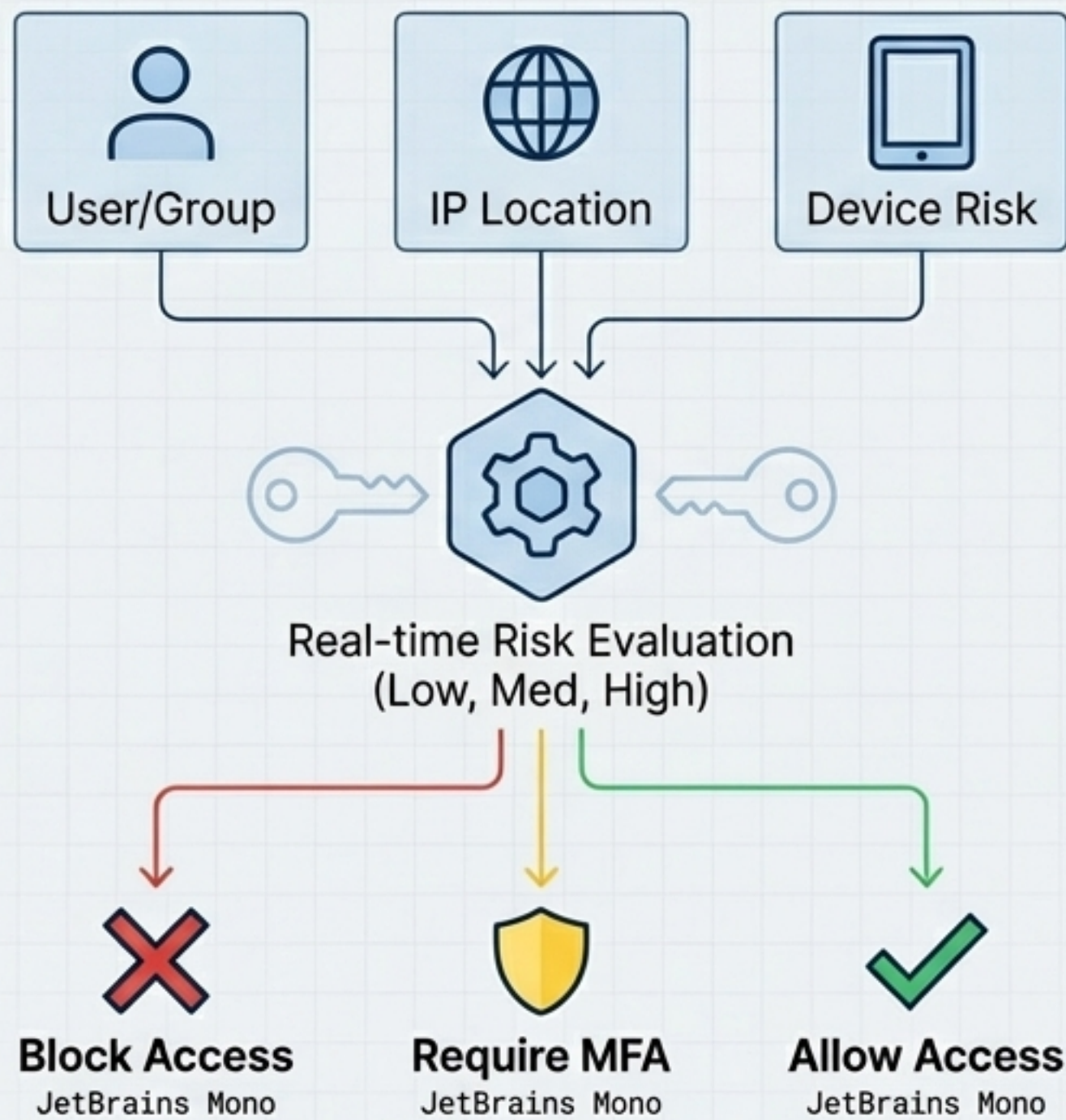
Hazard Warning: Prepare for diverse intelligence formats: Case studies, active screens, drag-and-drop lists, and hot area diagrams. Some question blocks cannot be skipped or revisited.

The Editorial Architect Blueprint

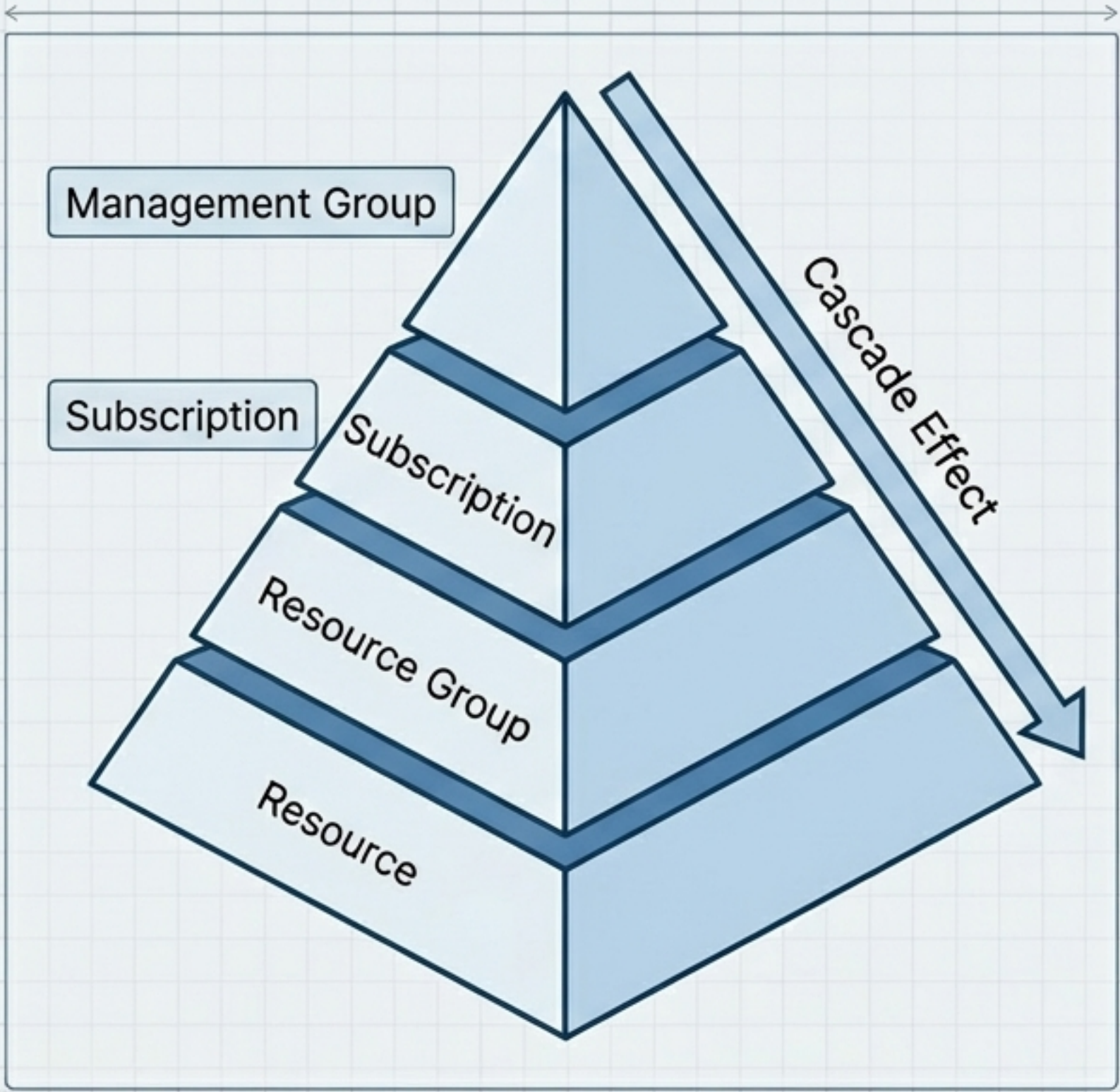
Identity Types



Conditional Access Flowchart



The Editorial Architect Blueprint



RBAC vs. Azure Policy

Dimension	RBAC	Azure Policy
Focus	User Actions (Who can do what?)	Resource Compliance (What can exist?)
Mechanism	Roles (Owner, Contributor, Reader)	Effects (Deny, Audit, Append)
Example	Alice can restart VMs.	VMs can only be deployed in the East US region.

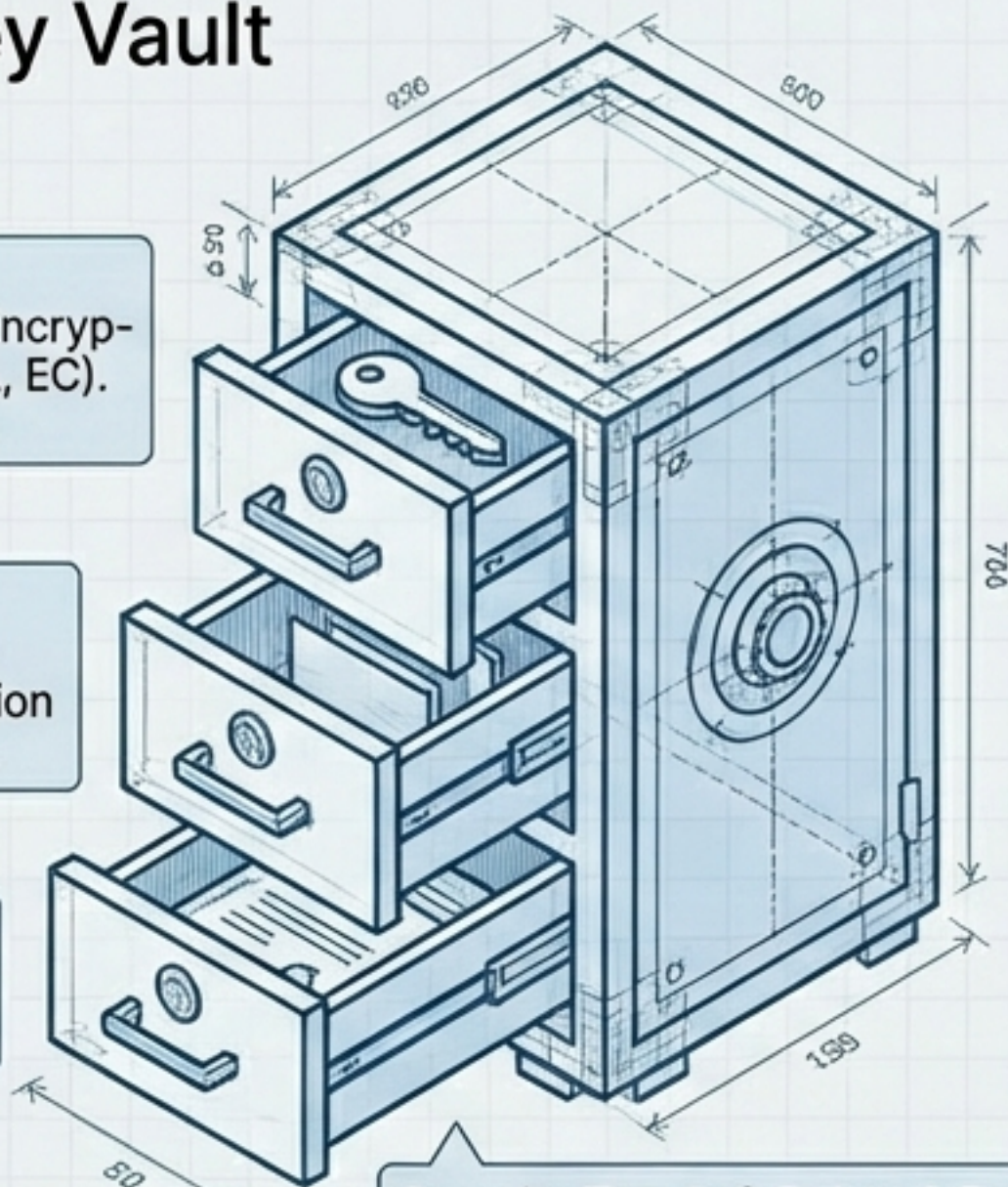
The Editorial Architect Blueprint

Azure Key Vault

Keys:
Cryptographic/Encryption assets (RSA, EC).
Supports HSMs.

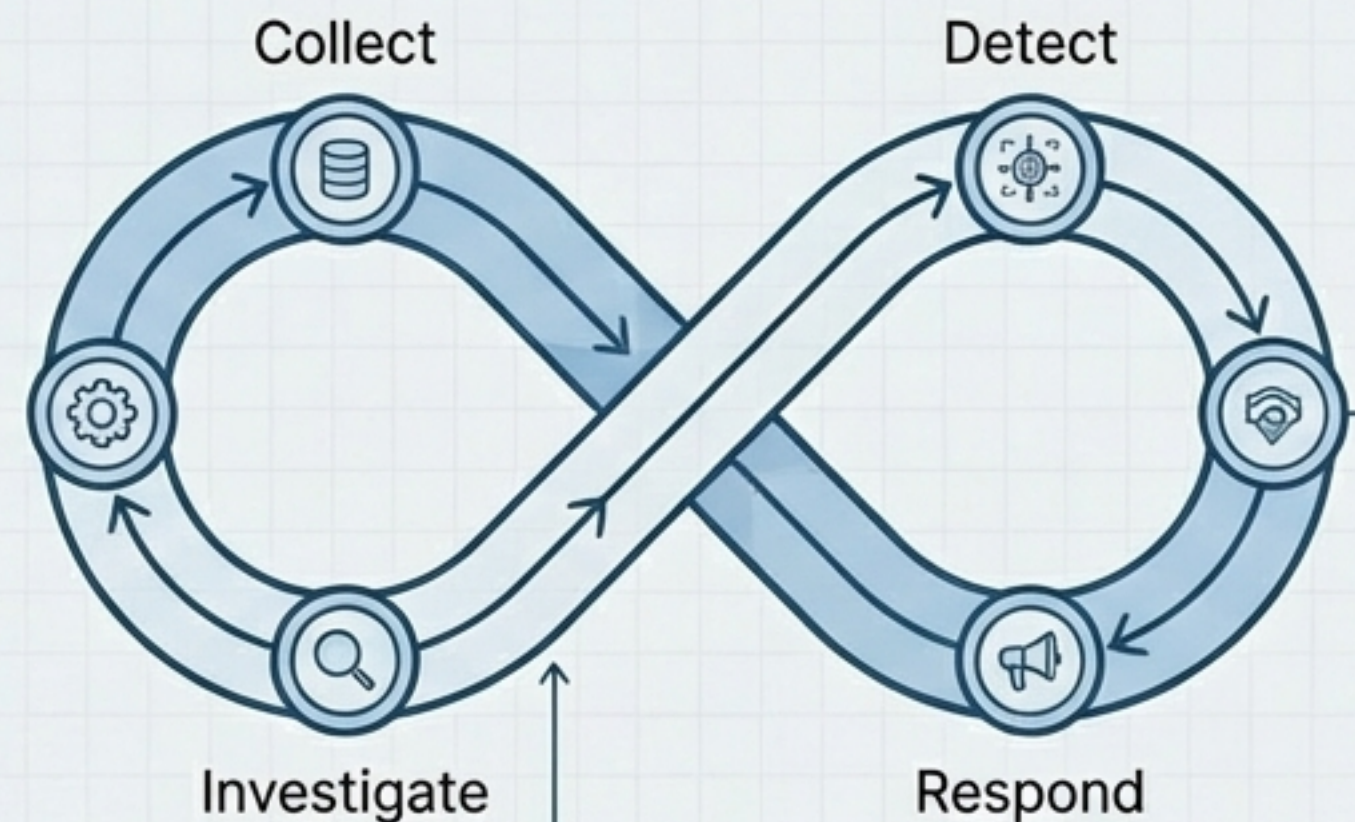
Secrets:
Passwords, API
tokens, connection
strings (<25KB).

Certificates:
x509, SSL/TLS
management.



Security Tag: Soft Delete & Purge
Protection prevent accidental or
malicious permanent data loss.

Microsoft Sentinel



SIEM (Analytics & Threat Intel)
seamlessly integrated with **SOAR**
(Automated response via Logic
Apps).

The Editorial Architect Blueprint

Inputs: Dynamic values provided at deployment.

Returned values: Data returned post-deployment (e.g., generated IP address).

```
{
  $schema: ..., "http://amla.u/schema",
  contentVersion: ...,
  parameters: { ... },
  variables: { ... }, ←
  functions: [ ... ], ←
  resources: [ ... ], ←
  outputs: { ... }
}
```

Custom logic: Reusable logic steps.

Computed values: Processed logic for internal template use.

Azure components: The actual infrastructure being built (VMs, networks).

Declarative, but unforgiving: If an ARM template deployment fails mid-execution, it does not roll back automatically. Orphaned resources remain.

The Editorial Architect Blueprint

Data Access Tiers

Hot

Highest storage cost, lowest access cost. For active data.

Cool

Lower storage cost, higher access cost. 30-day minimum retention.

Archive

Lowest storage cost, highest access cost. 180-day minimum. Requires rehydration (hours) to read.

Replication Durability Spectrum

LRS (Locally Redundant)
1 region, 1 data center.
(11 nines)

ZRS (Zone Redundant)
1 region, 3 availability
zones. (12 nines)

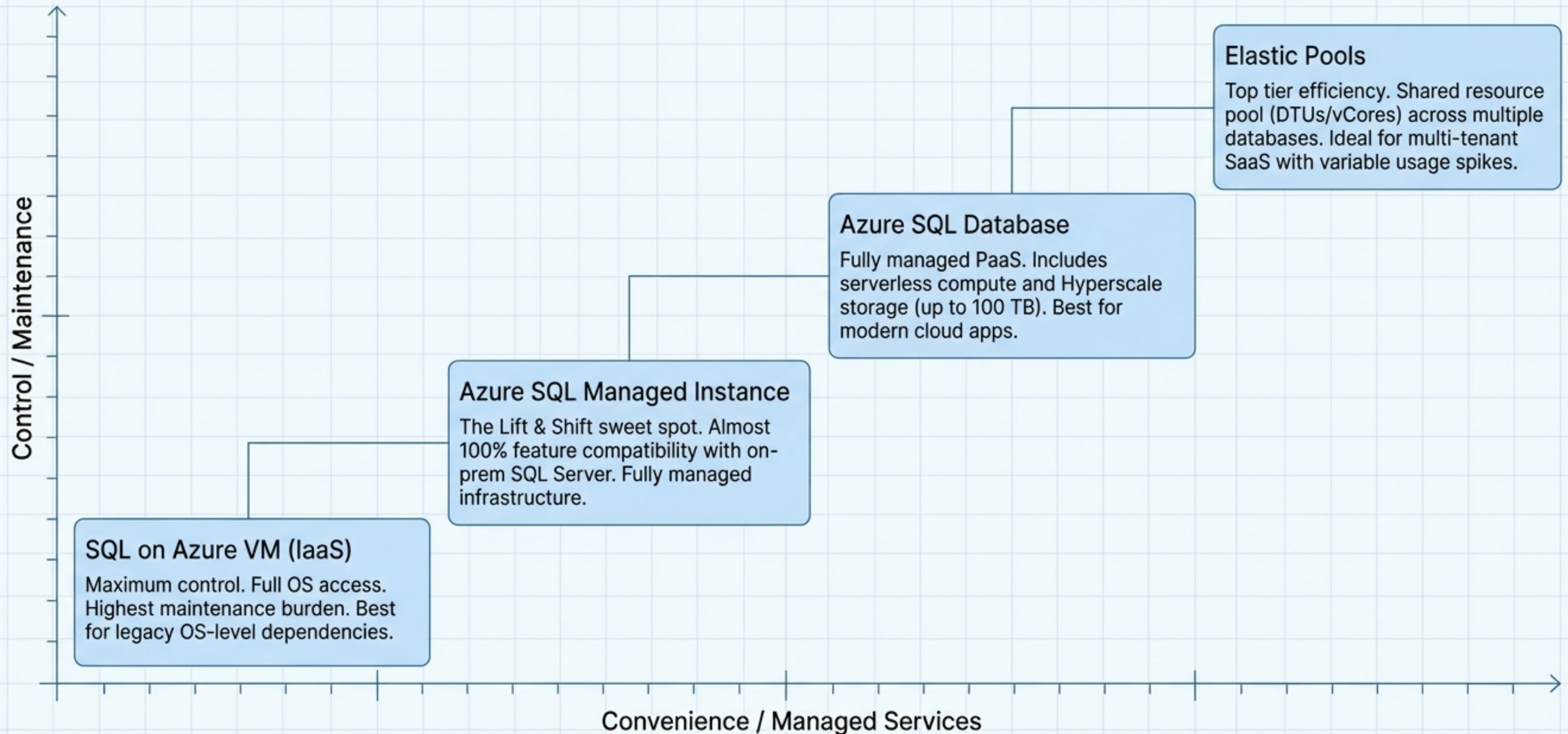
GRS (Geo-Redundant)
2 regions, async copy.
(16 nines)

GZRS (Geo-Zone Redundant)
2 regions, multi-zone
primary. (16 nines)

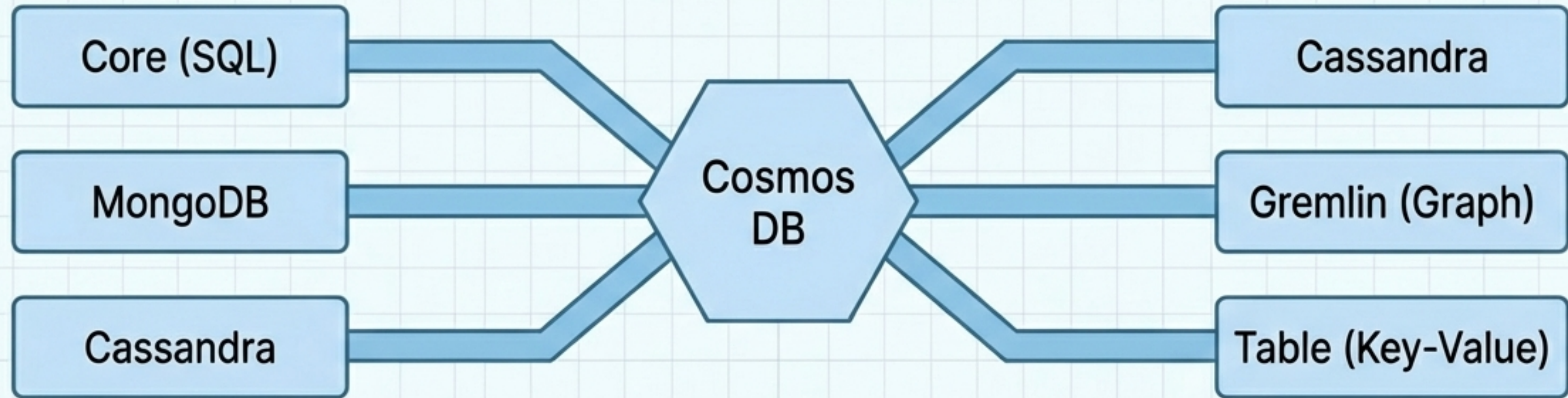


Increasing Cost & Durability

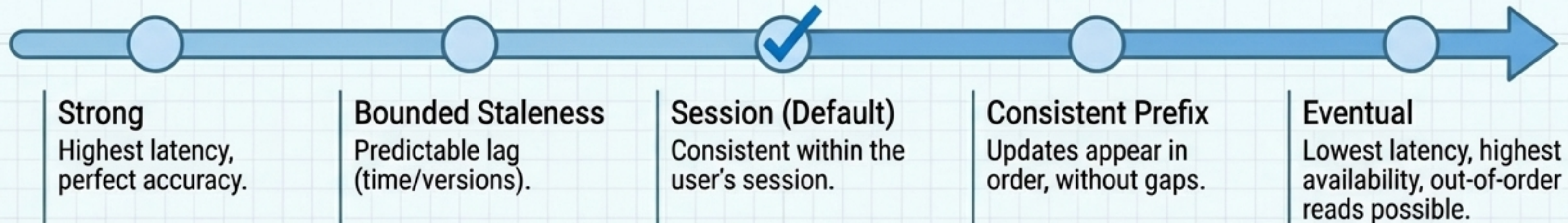
The Editorial Architect Blueprint



The Editorial Architect Blueprint



Consistency Spectrum



Architect Note: Data is distributed via Partition Keys. Logical partitions group identical keys; Physical partitions manage the underlying compute/storage.

The Editorial Architect Blueprint

1. Ingest (Azure Data Factory)

Over 95 native connectors.
Orchestrates ETL/ELT
pipelines.

2. Store (Data Lake Storage Gen 2)

Built on Blob storage with
a true hierarchical name-
space (directories/folders)
for massive analytical
workloads.

3. Prepare & Train (Databricks)

Apache Spark-based
environment for
collaborative data
engineering and machine
learning.

4. Serve & Analyze (Synapse Analytics)

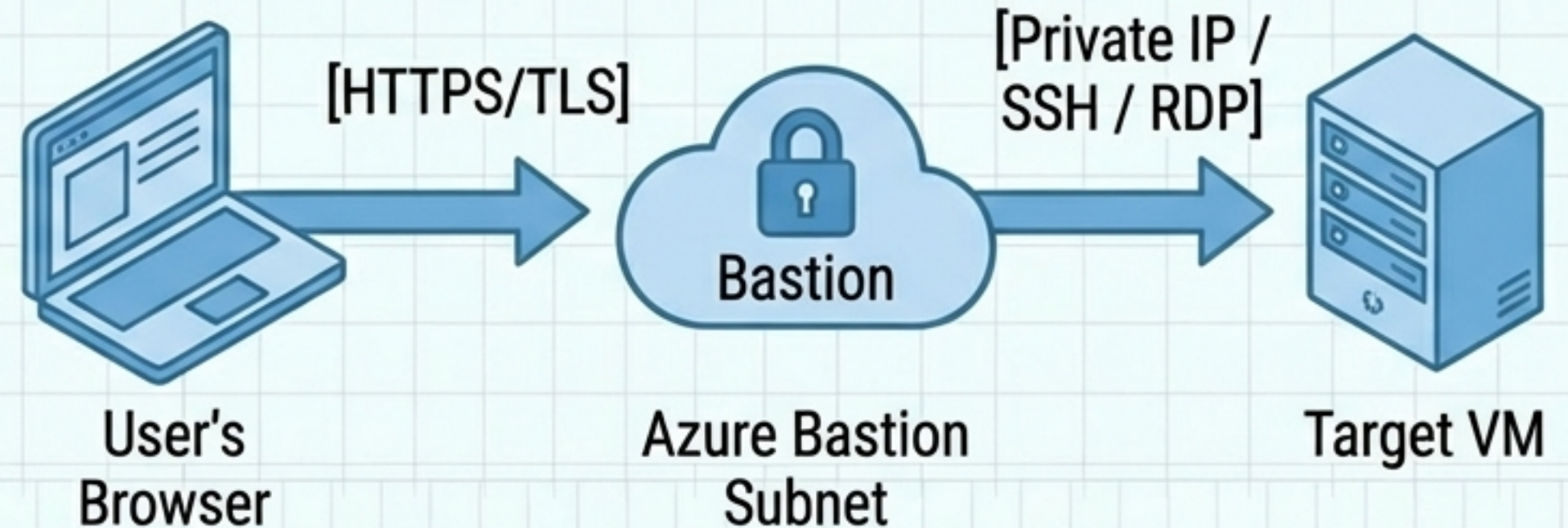
Unifies data warehousing.
Offers Dedicated SQL
Pools (predictable work-
loads) and Serverless SQL
Pools (unplanned, bursty
queries over the lake).

The Editorial Architect Blueprint

VM Sizing Decoded

- D** **[D-Series]:** General Purpose
(Balanced CPU/Memory)
- E** **[E-Series]:** Memory Optimized
(Large caches, in-memory DBs)
- F** **[F-Series]:** Compute Optimized
(Batch processing, web servers)
- L** **[L-Series]:** Storage Optimized
(High disk I/O)

Azure Bastion Architecture



Architect Note: Zero public IP exposure. Mitigates port scanning and RDP brute-force attacks.

The Editorial Architect Blueprint

AKS Division of Responsibility

Control Plane - Managed by Azure

Fully managed by Microsoft.
No cost for the master nodes.

API Server

Scheduler

etcd (key-value store)

Agent Nodes - Managed by Customer

Customer manages scaling and pays for
the underlying VM compute resources.

VM Scale Sets

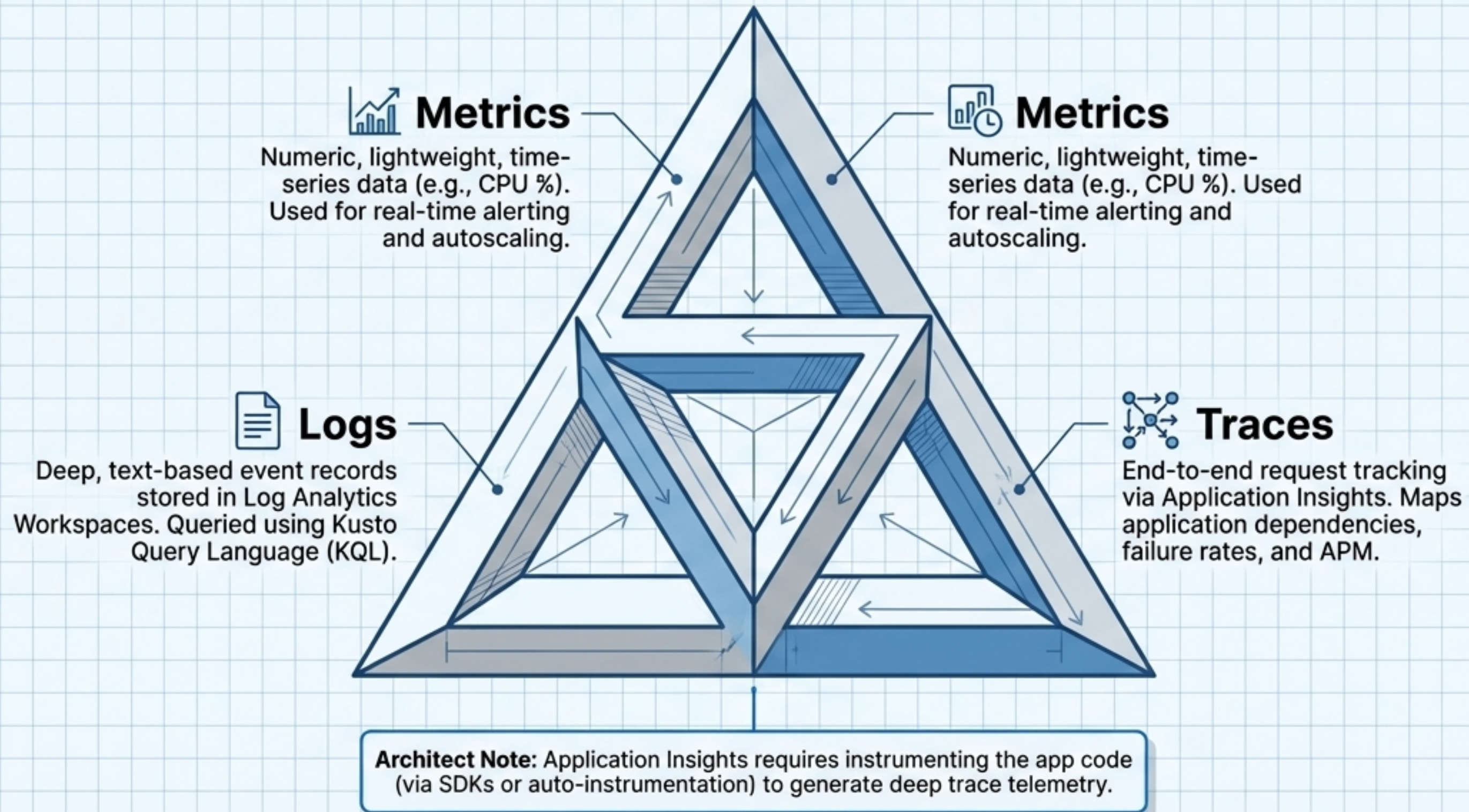
Kubelet & Container Runtimes

User Pods

Architect Note: Bridge to Kubernetes enables local developers to write and test microservice code directly against an active AKS cluster.

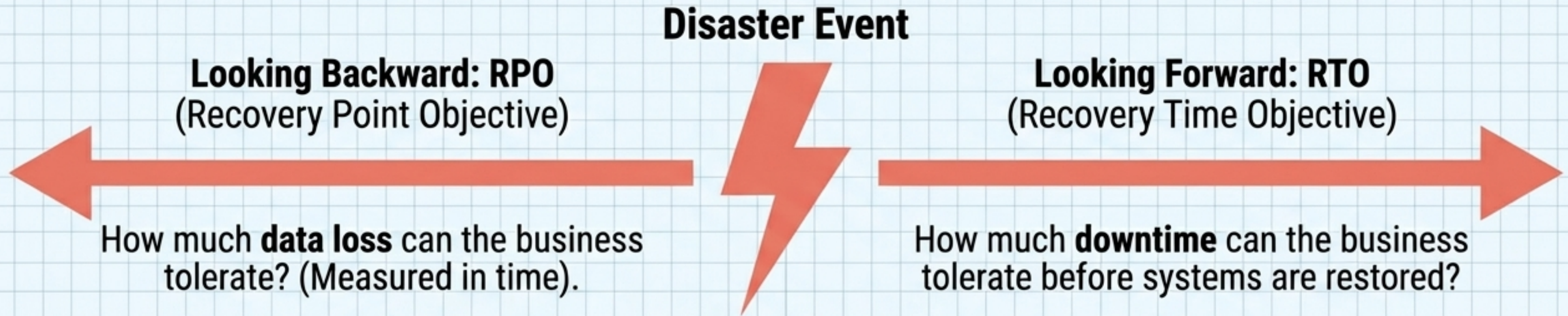
The Editorial Architect Blueprint

The Triforce of Observability (Azure Monitor)



The Editorial Architect Blueprint

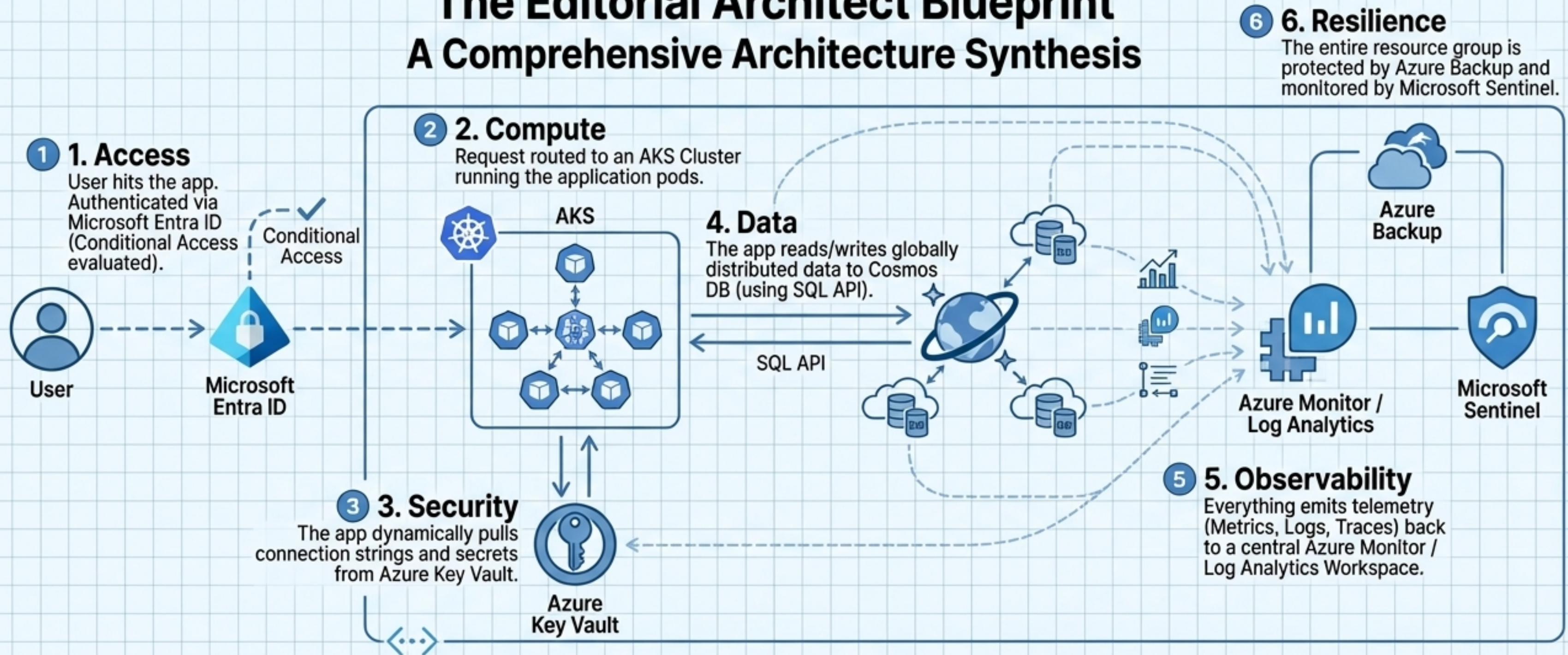
Disaster Recovery Concepts: RPO, RTO, and Azure Solutions



Azure Backup	Azure Site Recovery (ASR)
State protection. Safeguards historical data state (files, folders, DBs) against corruption or ransomware. Uses Recovery Services Vault.	Orchestration. Replicates workloads to a secondary region and orchestrates the failover networking to keep applications online.

Real-World AZ-305 Architecture Synthesis

The Editorial Architect Blueprint A Comprehensive Architecture Synthesis



*Architecture is not just choosing services;
it is designing how they communicate, scale, and survive.*